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Hoang Trieu Vy LE
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Hoang Trieu Vy LE

SUMMARY — I am a Senior Principal R&D Engineer at Beink, where I lead Beink 3D Studio, an AI-native 3D modeling tool with applications in product design, architecture and retail. My work centres on agent harness engineering: the orchestration, tools, memory, evaluation and cost control that let multimodal agents reason and build reliably in production. My research interests include 3D/4D scene understanding, robotics and human modelling. Before Beink, I was a Research Engineer at CEA. I completed my Ph.D. at École Normale Supérieure de Lyon under Prof. Nelly Pustelnik, working on unrolled proximal architectures that bridge signal processing and deep learning.

EXPERIENCES

Beink Dream **Mar 2025 – Present**
Senior Research Scientist *Paris, France*

- **Project Lead, Beink 3D Studio**: Led a new AI-native 3D modeling tool for architecture and product design from idea to production, turning sketches, images and plain language into editable 3D models.
- **AI architecture**: Defined the end-to-end architecture of the company's AI stack — multimodal model selection, multi-agent orchestration, sandboxed geometry engines, inference and serving, cost controls and evaluation harnesses — and took it from research prototype to production at scale.

CEA – DIGITEO Labs **Feb 2024 – Feb 2025**
Research Engineer *Saclay, France*

Lightweight learning models for 3D CT reconstruction — physics-consistent volumetric container representations recovered from sparse, noisy and limited high-dimensional measurements.

INMA, ICTEAM – UC Louvain **Sep 2020 – Aug 2021**
Visiting PhD Researcher *Louvain, Belgium*

Designed scalable accelerated algorithms to solve joint image restoration and edge detection on large-scale physics experiment images.

OGoxe – Météo France **Jan 2020 – Aug 2020**
Research Intern *Toulouse, France*

Built velocity estimation models from video sequences — inferring physical dynamics (fluid flow) from continuous visual observations, an early instance of physics-grounded world dynamics modelling.

IMFT – Toulouse Fluid Mechanics Institute **Jul 2019 – Oct 2019**
Research Intern – High Performance Computing *Toulouse, France*

Novel bucketing-based algorithm for particle-based simulation using mesh-less models across GPU clusters.

EDUCATION

École Normale Supérieure de Lyon **Sep 2020 – Dec 2023**
Ph.D. in Applied Mathematics, AI, Signal and Image Processing *Lyon, France*

Thesis: *Variations on the Mumford-Shah functional for interface detection in degraded images: from proximal algorithms to unrolled architectures.*
Advised by Prof. Nelly Pustelnik & Dr. Marion Foare.

INSA Toulouse **Sep 2015 – Sep 2020**
Master of Engineering in Applied Mathematics *Toulouse, France*

Major in Mathematical Modelling and Numerical Analysis.

PUBLICATIONS

Unfolded Discrete Mumford-Shah Functional for Joint Image Denoising and Edge Detection, ***EUSIPCO 2025***

H.T.V Le, Marion Foare, Audrey Repetti, Nelly Pustelnik

Embedding Blake-Zisserman Regularization in Unfolded Proximal Neural Networks for Enhanced Edge Detection, ***IEEE SPL 2025***

H.T.V Le, Marion Foare, Audrey Repetti, Nelly Pustelnik

Proximal Neural Networks based Reconstruction for Few-View CT Applications, ***iCT 2025***

H.T.V Le, C. Bossuyt, J. Escoda, H. Wang, J. De Beenhouwer, J. Sijbers

Unfolded Proximal Neural Networks for Robust Image Gaussian Denoising, ***IEEE TIP 2024***

H.T.V Le, Audrey Repetti, Nelly Pustelnik

Deep Image Prior for Sparse-View Reconstruction in Static, Rectangular Multi-Source X-ray CT Systems for Cargo Scanning, ***SPIE Developments in X-Ray Tomography XV, 2024***

C. Bossuyt, J. De Beenhouwer, J. Sijbers, D. Iuso, M. Costin, J. Escoda, H.T.V Le, A. J. den Dekker

Proximal Based Strategies for Solving Discrete Mumford-Shah With Ambrosio-Tortorelli Penalization on Edges, ***IEEE SPL 2022***

H.T.V Le, Marion Foare, Nelly Pustelnik

The Faster Proximal Algorithm, the Better Unfolded Deep Learning Architecture? A Study Case of Image Denoising, ***EUSIPCO 2022***

H.T.V Le, Nelly Pustelnik, Marion Foare

Fast Unfolded Proximal Algorithms for the Analysis of Piecewise Homogeneous Fractal Images, ***Gretsi 2022***

H.T.V Le, Barbara Pascal, Nelly Pustelnik, Marion Foare, Patrice Abry

ACTIVITIES

Teaching: Signal processing, convex optimization, unfolded NNs and inverse problems (ENS Lyon); Signal processing and Cryptography Mathematics (EPITA Lyon); Fundamental Machine Learning (EPITA Paris)

Talks:

EUSIPCO, Belgrade, Serbia **Aug 2022**

Conference on Digital Geometry and Discrete Variational Calculus **Mar 2021**

Projects & Open-Source Software:

Zingu [Python, Embedded] — Two-wheeled self-balancing robot that recovers autonomously after a fall; ongoing work on the perception and decision stack.
Proximal Neural Networks [PyTorch] — Lightweight unrolled architectures combining iterative algorithms and deep networks for inverse problems.
P-DMS-AT [Python] — Proximal strategies for Discrete Mumford-Shah with Ambrosio-Tortorelli edge penalization.
BZ-PNN [PyTorch] — Super-lightweight model for enhanced edge detection via Blake-Zisserman regularization.
DMS-PNN [PyTorch] — Unfolded Discrete-Mumford-Shah Proximal Neural Networks.
Bucketing-based Algorithm [Fortran, C/C++, Python] — Parallelized linearly transformed particle method for diffusion equations.

AWARDS

- Third prize, Vietnam National Physics Olympiad **2015**
- Fourth prize, Vietnam National Physics Olympiad **2014**
- Silver medal, Northern Regional Vietnam Physics Olympiad **2013**